

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

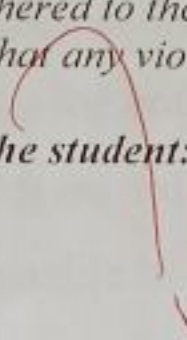
Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I Shaik. Imran (192525348) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



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ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

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	Problem understanding (20%)	method selection (20%)	solution process (25%)	Accuracy of calculation (20%)	Interpretation Results (15%)
Q1	1	2	2	1.5	1 7.5
Q2	1	1.5	1.5	2	1 7
Q3	1.5	1	2	1	1.5 7
Q4	2	1.5	2.5	1.5	1.5 9
Q5	1.5	2	2	2	1 8.5

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1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

Answer

Given IP address block: **192.168.20.0/24**

The network is divided into three subnets:

- **Lab A:** /26
- **Lab B:** /27
- **Lab C:** /28

Assuming the subnets are allocated sequentially from **192.168.20.0**, the subnet details are as follows.

1. Lab A – /26 Subnet

Subnet Mask: 255.255.255.192

Block Size: 64 addresses

- **Network Address:** 192.168.20.0
- **First Host Address:** 192.168.20.1
- **Last Host Address:** 192.168.20.62
- **Broadcast Address:** 192.168.20.63
- **Total Usable Host IP Addresses:** 62

2. Lab B – /27 Subnet

The next available address after Lab A is **192.168.20.64**.

Subnet Mask: 255.255.255.224

Block Size: 32 addresses

- **Network Address:** 192.168.20.64
- **First Host Address:** 192.168.20.65
- **Last Host Address:** 192.168.20.94
- **Broadcast Address:** 192.168.20.95
- **Total Usable Host IP Addresses:** 30

3. Lab C – /28 Subnet

The next available address after Lab B is **192.168.20.96**.

Subnet Mask: 255.255.255.240

Block Size: 16 addresses

- **Network Address:** 192.168.20.96
- **First Host Address:** 192.168.20.97
- **Last Host Address:** 192.168.20.110
- **Broadcast Address:** 192.168.20.111
- **Total Usable Host IP Addresses:** 14

Final Answer Table

Lab	Subnet	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
Lab A	/26	192.168.20.0	192.168.20.1 – 192.168.20.62	192.168.20.63	62
Lab B	/27	192.168.20.64	192.168.20.65 – 192.168.20.94	192.168.20.95	30
Lab C	/28	192.168.20.96	192.168.20.97 – 192.168.20.110	192.168.20.111	14

Calculation of Usable Hosts

Usable Hosts = $2^{(32 - \text{Prefix Length})} - 2$

- /26: $2^6 - 2 = 62$ hosts
- /27: $2^5 - 2 = 30$ hosts
- /28: $2^4 - 2 = 14$ hosts

Therefore, Lab A has 62 usable IP addresses, Lab B has 30 usable IP addresses, and Lab C has 14 usable IP addresses.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

Answer

Given:

System 1

- IP Address: **192.168.1.130**
- Subnet Mask: **255.255.255.192**
- CIDR Notation: **/26**

System 2

- IP Address: **192.168.1.190**
- Subnet Mask: **255.255.255.192**
- CIDR Notation: **/26**

Step 1: Find the Subnet Block Size

Subnet mask **255.255.255.192** means **/26**.

Block Size:

$$256 - 192 = 64$$

Therefore, the subnet ranges are:

- 192.168.1.0 – 192.168.1.63
- 192.168.1.64 – 192.168.1.127
- 192.168.1.128 – 192.168.1.191
- 192.168.1.192 – 192.168.1.255

System 1: 192.168.1.130/26

The IP address **192.168.1.130** falls in the range:

192.168.1.128 – 192.168.1.191

Therefore:

- **Network Address:** 192.168.1.128
- **Broadcast Address:** 192.168.1.191
- **Valid Host Range:** 192.168.1.129 – 192.168.1.190
- **Given IP Address:** 192.168.1.130

Since **192.168.1.130** is between **192.168.1.129** and **192.168.1.190**, it is a **valid host IP address** in this subnet.

System 2: 192.168.1.190/26

The IP address **192.168.1.190** also falls in the range:

192.168.1.128 – 192.168.1.191

Therefore:

- **Network Address:** 192.168.1.128
- **Broadcast Address:** 192.168.1.191
- **Valid Host Range:** 192.168.1.129 – 192.168.1.190
- **Given IP Address:** 192.168.1.190

Since **192.168.1.190** is between **192.168.1.129** and **192.168.1.190**, it is also a **valid host IP address** in this subnet.

Number of Usable Host Addresses

For a /26 subnet:

$$\text{Usable Hosts} = 2^{(32 - 26)} - 2$$

$$= 2^6 - 2$$

$$= 64 - 2$$

$$= 62 \text{ usable host addresses}$$

Therefore, each /26 subnet has **62 usable host IP addresses**.

Final Answer Table

System	IP Address	Subnet Mask	Network Address	Subnet Range	Valid Host Range	Broadcast Address	Valid ?	Usable Hosts
System 1	192.168.1.130	255.255.255.192 (/26)	192.168.1.128	192.168.1.128 – 192.168.1.191	192.168.1.129 – 192.168.1.190	192.168.1.191	Yes	62
System 2	192.168.1.190	255.255.255.192 (/26)	192.168.1.128	192.168.1.128 – 192.168.1.191	192.168.1.129 – 192.168.1.190	192.168.1.191	Yes	62

Conclusion

Both **192.168.1.130** and **192.168.1.190** belong to the **same subnet: 192.168.1.128/26**. Both IP addresses are valid host addresses, and the subnet provides **62 usable host IP addresses**.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Given IP Block: 172.16.0.0/16

The organization has four departments with the following subnet masks:

- **HR** → /25
- **Sales** → /26
- **IT** → /27
- **Admin** → /28

Assuming the subnets are allocated **sequentially**, starting from **172.16.0.0**, the allocation is as follows.

HR Department – /25

Subnet Mask: 255.255.255.128

Total Addresses: 128

- **Network Address:** 172.16.0.0
- **First Host:** 172.16.0.1
- **Last Host:** 172.16.0.126
- **Broadcast Address:** 172.16.0.127
- **Usable Host Addresses:** 126

Sales Department – /26

The next available subnet starts at **172.16.0.128**.

Subnet Mask: 255.255.255.192

Total Addresses: 64

- **Network Address:** 172.16.0.128
- **First Host:** 172.16.0.129
- **Last Host:** 172.16.0.190
- **Broadcast Address:** 172.16.0.191
- **Usable Host Addresses:** 62

IT Department – /27

The next available subnet starts at **172.16.0.192**.

Subnet Mask: 255.255.255.224

Total Addresses: 32

- **Network Address:** 172.16.0.192
- **First Host:** 172.16.0.193
- **Last Host:** 172.16.0.222
- **Broadcast Address:** 172.16.0.223
- **Usable Host Addresses:** 30

Admin Department – /28

The next available subnet starts at **172.16.0.224**.

Subnet Mask: 255.255.255.240

Total Addresses: 16

- **Network Address:** 172.16.0.224
- **First Host:** 172.16.0.225
- **Last Host:** 172.16.0.238
- **Broadcast Address:** 172.16.0.239
- **Usable Host Addresses:** 14

Final Answer Table

Department	Subnet Mask	Network Address	Valid Host IP Range	Broadcast Address	Usable Hosts
HR	/25	172.16.0.0	172.16.0.1 – 172.16.0.126	172.16.0.127	126
Sales	/26	172.16.0.128	172.16.0.129 – 172.16.0.190	172.16.0.191	62
IT	/27	172.16.0.192	172.16.0.193 – 172.16.0.222	172.16.0.223	30
Admin	/28	172.16.0.224	172.16.0.225 – 172.16.0.238	172.16.0.239	14

Usable Host Calculation

Usable Hosts = $2^{(32 - \text{Prefix Length})} - 2$

- **HR /25:** $2^7 - 2 = 126$
- **Sales /26:** $2^6 - 2 = 62$
- **IT /27:** $2^5 - 2 = 30$
- **Admin /28:** $2^4 - 2 = 14$

Conclusion

The four departments are successfully allocated separate subnets from the **172.16.0.0/16** network. HR receives **126 usable hosts**, Sales receives **62**, IT receives **30**, and Admin receives **14 usable host addresses**.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

Given

- **Network Address:** 172.16.10.0/24
- **New Subnet Mask:** /27
- **IP Address 1:** 172.16.10.78
- **IP Address 2:** 172.16.10.95

Total Number of Subnets Created

The original network is /24 and it is subnetted using /27.

Number of borrowed bits = $27 - 24 = 3$

Therefore:

Number of Subnets = $2^3 = 8$ subnets

The 8 Subnets are:

1. 172.16.10.0 – 172.16.10.31
2. 172.16.10.32 – 172.16.10.63
3. 172.16.10.64 – 172.16.10.95
4. 172.16.10.96 – 172.16.10.127
5. 172.16.10.128 – 172.16.10.159
6. 172.16.10.160 – 172.16.10.191
7. 172.16.10.192 – 172.16.10.223
8. 172.16.10.224 – 172.16.10.255

Number of Usable Hosts per Subnet

For a /27 subnet:

Host Bits = $32 - 27 = 5$

Total Addresses = $2^5 = 32$

Usable host addresses:

$32 - 2 = 30$

Therefore, **each subnet has 30 usable host addresses.**

Subnet of IP Address 172.16.10.78

A /27 subnet has a block size of:

$$256 - 224 = 32$$

The subnet ranges increase by 32:

- 0–31
- 32–63
- **64–95**
- 96–127

The IP address **172.16.10.78** falls in the range **172.16.10.64 – 172.16.10.95**.

Therefore:

- **Network Address:** 172.16.10.64
- **First Host Address:** 172.16.10.65
- **Last Host Address:** 172.16.10.94
- **Broadcast Address:** 172.16.10.95

So, **172.16.10.78** belongs to the subnet **172.16.10.64/27**.

Check Whether 172.16.10.95 Is in the Same Subnet

The subnet range is:

$$172.16.10.64 - 172.16.10.95$$

Therefore, **172.16.10.95** belongs to the same subnet range.

However, **172.16.10.95** is the broadcast address, so it cannot be assigned to a host.

Final Answer Table

Item	Result
Original Network	172.16.10.0/24
Subnet Mask	/27 (255.255.255.224)
Total Number of Subnets	8
Usable Hosts per Subnet	30
IP Address	172.16.10.78
Subnet	172.16.10.64/27
Network Address	172.16.10.64
Valid Host Range	172.16.10.65 – 172.16.10.94
Broadcast Address	172.16.10.95

Is 172.16.10.95 in the Same Subnet? **Yes**

Can 172.16.10.95 be used as a Host? **No, it is the broadcast address**

Conclusion

The **172.16.10.0/24** network is divided into **8 subnets** using a **/27 subnet mask**. Each subnet contains **30 usable host addresses**. The IP address **172.16.10.78** belongs to the **172.16.10.64/27 subnet**, and **172.16.10.95** is also within the same subnet range, but it is the **broadcast address**.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

Answer

Given IPv6 Address

2001:0db8:0000:0000:0000:ff00:0042:8329

Each IPv6 address contains **8 groups**, and each group normally has **4 hexadecimal digits**.

Compress the IPv6 Address

IPv6 compression rules:

1. Remove leading zeros from each group.
2. Replace one continuous sequence of all-zero groups with **::**.

Given address:

2001:0db8:0000:0000:0000:ff00:0042:8329

After removing leading zeros:

2001:db8:0:0:0:ff00:42:8329

There are three consecutive groups containing **0000**, so they can be replaced with **::**.

Compressed Address:

2001:db8::ff00:42:8329

Expand the Compressed Address

Compressed form:

2001:db8::ff00:42:8329

The :: represents the missing zero groups required to make a total of 8 groups.

Expanded Address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Thus, the expanded address is the same as the original full IPv6 address.

Determine the Network Prefix Using /64

An IPv6 prefix length of /64 means that the first **64 bits** represent the network portion.

Since each group contains **16 bits**:

$$64 \div 16 = 4 \text{ groups}$$

Therefore, the first four groups form the network prefix:

2001:0db8:0000:0000

Network Prefix:

2001:db8:0:0::/64

It can also be written as:

2001:db8::/64

Calculate the Number of Possible Host Addresses

IPv6 address size = **128 bits**

Prefix length = **64 bits**

Therefore:

$$\text{Host bits} = 128 - 64 = 64 \text{ bits}$$

Total possible host addresses:

$$2^{64}$$

$$= 18,446,744,073,709,551,616 \text{ possible addresses}$$

Final Answer Table

Item	Answer
Given IPv6 Address	2001:0db8:0000:0000:0000:ff00:0042:8329

Item	Answer
Compressed Form	2001:db8::ff00:42:8329
Expanded Form	2001:0db8:0000:0000:0000:ff00:0042:8329
Prefix Length	/64
Network Prefix	2001:db8::/64
Host Bits	64 bits
Total Possible Host Addresses	2⁶⁴
Number of Host Addresses	18,446,744,073,709,551,616

Conclusion

The IPv6 address **2001:0db8:0000:0000:0000:ff00:0042:8329** is compressed as **2001:db8::ff00:42:8329**. With a **/64 prefix**, the network prefix is **2001:db8::/64**, leaving **64 bits for host addresses**. Therefore, the network can support **2⁶⁴ possible host addresses**.

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